- Cloud native applications are horizontally scalable, loosely coupled, highly cohesive, resilient to faults, manageable, and observable.
- Cloud native development is supported by automation, continuous delivery, and DevOps.
- Continuous delivery is a holistic engineering practice for delivering high-quality software quickly, reliably, and safely.
- DevOps is a culture enabling collaboration among different roles to deliver business value together.
- Modern businesses go cloud native to produce software that can be delivered quickly, can be scaled dynamically depending on demand, and is always available and resilient to failures while optimizing costs.
- Containers (such as Docker containers) can be used as computational units when designing cloud native systems. They are more lightweight than virtual machines and provide portability, immutability, and flexibility.
- Dedicated platforms (such as Kubernetes) offer services to manage containers without directly handling the underlying layers. They provide container orchestration, cluster management, network services, and scheduling.
- Serverless computing is a model where the platform (such as Knative) manages servers and the underlying infrastructure, and the developer only focuses on the business logic. The backend functionality is enabled on a pay-per-use basis for cost optimization.
- A microservices architecture can be used to build cloud native applications, but it’s not a requirement.